

Ivan Reshetnikov

Machine Learning Engineer & AI Engineer

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Professional Summary

ML engineer with 5+ years shipping models to production and building LLM workflows since 2023. At a B2B marketplace with 15M supplier offers, I built a matching pipeline that halved the offers hidden from customers at over 99% precision.

As an AI engineer, I build LLM applications with evaluation, guardrails, and human-in-the-loop controls. I take projects end-to-end, from the business cost of errors to deployment and monitoring in production.

Work Experience

Senior Machine Learning Engineer

Maksmart

Aug 2024 - Apr 2026

Production ML and LLM systems for a B2B marketplace: 3M product cards, 15M supplier offers, 300K offer updates a day. Shipped 8 models to production across matching, catalog quality, pricing, and forecasting.

- Halved the number of supplier offers hidden from customers, matching them to product cards at precision above 99% with fastText retrieval and LightGBM reranking.
- Cut non-obvious duplicate product cards by 60% at 95% precision, merging them with an attention-based PyTorch scorer and graph-based grouping.
- Assigned product types to several million product cards using metric learning and LLM + RAG labeling, enabling grouping beyond existing catalog categories.
- Scored every incoming supplier offer for price anomalies so moderators reviewed the riskiest first, using a LightGBM ensemble with uncertainty estimates.
- Replaced fixed delivery estimates with data-driven targets for carriers, including routes without shipment history, while reducing a combined score of error and bias by 28%.

Core technologies: Python, PyTorch, LightGBM, scikit-learn, fastText, Elasticsearch, LLM + RAG, FastAPI, Docker, Kubernetes, PostgreSQL, ClickHouse.

Data Scientist

Architech NYC

Apr 2022 - Jul 2024

Only developer on each ML and LLM project for a container transportation company, working with the owner and dispatchers from requirements to production: rate calculation, email handling, document checks, and reporting.

- Built a rate suggestion service that dispatchers used for customer quotes, at 11% RMSPE on future periods, with daily retraining gated by validation.
- Sped up email replies with an LLM assistant that pulled shipment data and calculator rates, sending status updates automatically and drafting quotes for review.
- Automated driver document validation with OCR, signature detection, and cross-checks against internal shipment records, sending uncertain cases to dispatcher review.
- Let the owner track operations with Apache Superset dashboards, from late drivers and container yard time to order volumes and sales rep performance.

Machine Learning Engineer

Sape

Dec 2020 - Mar 2022

ML for SEO, search, and ad exchange products, with three models shipped to production.

- Replaced tag-based and random backlink discovery with embedding-based semantic search that got positive user feedback and further company investment.
- Shipped the search over 48M pages as a Docker service with an HTTP API that the backlink marketplace team deployed and ran in their product.
- Assigned topics to publisher sites automatically across 68 categories, using fastText embeddings and logistic regression served through FastAPI.
- Replaced a model that recommended the minimum price for 55% of publisher sites with a combined site score and a price curve fitted on past deals.

Ruby on Rails Developer

Sape

Dec 2016 - Nov 2020

Backend for a self-service ad platform running campaigns across Facebook, Google Ads, VK, and myTarget.

- Kept campaign statuses and spend in sync with each ad network's API so that client balances were charged correctly.
- Unified the API for external users and the frontend, removing duplicated integration logic.
- Moved CI and hosting from a compromised Jenkins server to self-hosted GitLab with configuration in the repository.

Earlier Experience

- **Ruby on Rails Developer**, ORDIN, Apr 2016 - Nov 2016: Rails apps for real estate and qualification workflows.
- **Software Engineer**, Nasca LTD, Dec 2012 - Sep 2013 and Apr 2015 - Jan 2016: C++ trading systems.

Core Skills

Machine Learning: Python, PyTorch, LightGBM, scikit-learn, gradient boosting, NLP, embeddings, fastText, metric learning, active learning, anomaly detection, forecasting, uncertainty estimation, model evaluation

Search, Matching & Ranking: information retrieval, semantic search, vector search, candidate generation, reranking, product matching, duplicate detection, Elasticsearch, FAISS

LLM & RAG: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), agent workflows, tool calling, MCP, structured output, prompt engineering, LLM evaluation, guardrails, human-in-the-loop, OpenAI API, Claude Code

Production & Engineering: FastAPI, Docker, Kubernetes, Google Cloud Platform (GCP), CI/CD, model deployment, model monitoring, scheduled retraining, Airflow, PostgreSQL, ClickHouse, SQL, Pandas, Sentry

Certifications

Claude Certified Architect – Foundations by Anthropic (Aug 2026)

Education

Pskov State University

Master's degree, Information Systems and Technologies

2008 - 2013

Languages

English (*Professional working proficiency*), Russian (*Native*)